Pregnancy Support System

R26-IT-146

Project Proposal Report

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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Cardiotocography (CTG) scans are necessary for pregnant women to track uterine contractions and fetal heart rate trends. However, clinicians have historically interpreted CTGs by hand, which frequently causes delays, conflicting interpretations, accessibility issues for patients in remote areas, and a restricted capacity for continuous fetal monitoring. These difficulties draw attention to a serious weakness in the prenatal care systems in place today, which mainly rely on manual analysis, and in-person hospital stays. The present study proposes an innovative application for pregnant women based on an AI-assisted CTG Interpretation Module. This application allows users to input the values manually or upload images of the CTG report obtained through their device camera. This application uses an artificial intelligence algorithm for the analysis of the fetus's heart rate variability, accelerations, decelerations, and contractions to detect any symptoms of distress or abnormalities. To make the application highly accurate and reliable, the proposed module incorporates a hybrid model based on image processing and artificial intelligence for the analysis of CTG graph patterns and generation of relevant insights. The proposed application would provide pregnant women and their doctor with an overall report regarding the CTG scan results, which would include the conditions present, their possible reasons, severity levels, and preventive measures. This would help the doctor and the pregnant woman take immediate action regarding their health and provide better outcomes for the fetus. Faster detection of complications, ease of use for remote patients, reduced reliance on hospital visits, and timely detection of complications would be the key outcomes of the proposed application for pregnant women.

Keywords: CTG, Pregnancy Monitoring, AI Healthcare, Fetal Heart Rate Analysis, Medical Image Processing, Risk Prediction

ACKNOWLEDGEMENT

I am deeply grateful to my supervisor for the continuous guidance, encouragement, and invaluable support provided throughout the development of this research project. Their constructive feedback and expert insights significantly contributed to shaping the direction and quality of my work.

I would also like to extend my sincere appreciation to my teammates for their cooperation, commitment, and teamwork in creating the core components of our pregnancy-care mobile application. Their dedication and collaborative spirit greatly supported the successful integration of each module, including user health monitoring, AI-driven features, and doctor–patient interaction flows. My heartfelt thanks go to the Department of Information Technology at SLIIT for providing the essential facilities, academic resources, and technical support necessary to carry out this research. The environment and guidance offered by the department were instrumental in completing this project. Finally, I wish to acknowledge the medical professionals who assisted by sharing practical knowledge about CTG usage, maternal health concerns, and clinical considerations. Their real-world experience helped greatly in understanding the importance of accurate CTG interpretation and prenatal risk assessment, which strengthened the foundation of this research.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence
CTG – Cardiotocography
FHR – Fetal Heart Rate
ML – Machine Learning
API – Application Programming Interface
UI – User Interface
DB – Database
OCR – Optical Character Recognition
HCI – Human–Computer Interaction
NFR – Non-Functional Requirement

1. INTRODUCTION

1.1 Background and Context

Cardiotocography (CTG) is a widely used clinical method for monitoring fetal well-being during pregnancy. It records fetal heart rate patterns and uterine contractions, helping identify early signs of fetal distress, placental insufficiency, or abnormal contraction behavior. Timely CTG interpretation is critical because delayed detection of abnormal patterns can lead to complications such as fetal hypoxia, emergency delivery, or pregnancy loss.

However, in many regions, CTG interpretation remains a **manual, hospital-dependent, and time-consuming** process. Pregnant women—especially those living far from medical facilitie face difficulties accessing CTG analysis regularly. Additionally, overstretched healthcare professionals must interpret numerous CTG reports daily, increasing the likelihood of delays and inconsistencies.

Recent studies highlight that **AI-assisted medical diagnostics can significantly improve accuracy, speed, and accessibility**, especially for populations with limited access to specialists. In this context, an automated, mobile-based CTG interpretation system can bridge an important gap in maternal healthcare by enabling remote, reliable, and fast fetal monitoring.

Analysis of Existing Solutions and Their Limitations

Existing pregnancy mobile applications focus primarily on:

- Tracking symptoms
- Recording doctor visits
- Monitoring weight, sleep, and nutrition
- Providing educational information

However, **none of the widely available apps can analyze CTG graphs or interpret fetal heart rate patterns**. Similarly:

- No solution allows mothers to upload CTG images for AI interpretation
- Most systems rely entirely on in-hospital CTG machines and in-person doctor reviews
- There is no integrated doctor-feedback loop for remote CTG analysis
- There is limited automated risk prediction to alert mothers of possible complications

These limitations lead to delayed detection of fetal abnormalities and reduced accessibility for pregnant women who cannot regularly visit hospitals.

Ideas for the Proposed Solution and Its Parts

To address these issues, the proposed system introduces an **AI-powered CTG Interpretation Module**, a key component of the overall pregnancy monitoring app. The system:

- Allows mothers to manually enter CTG values or upload CTG report images

- Uses AI to interpret fetal heart rate variability, accelerations, decelerations, and contraction patterns
- Predicts fetal well-being using clinically relevant parameters
- Generate summary reports for both the mother and doctor
- Enables the doctor to review, amend, and send final comments

This approach integrates AI-assistance with medical decision-support, enabling faster, accurate, and more accessible CTG monitoring without relying on IoT machines.

1.2 Literature Review

Research on AI-assisted maternal health monitoring has grown significantly. Recent scholarly work demonstrates that machine learning techniques can identify fetal distress by analyzing CTG signals. Prior studies have used neural networks, decision trees, and statistical models to classify CTG readings with moderate to high accuracy.

However, existing research systems often suffer from limitations:

- Most models are tested in laboratory conditions, not deployed in real-world mobile applications
- AI models require high-quality digital CTG data, whereas mothers usually receive printed CTG reports
- Many studies do not incorporate image-based CTG interpretation
- Existing clinical systems lack integration with patient-facing mobile platforms

Commercial products are even more limited. Current pregnancy apps do not include automated CTG analysis, nor do they provide structured risk-level interpretation based on medical guidelines.

These gaps reinforce the need for a **practical, mobile-based, AI-driven CTG interpretation system** that supports doctors as well as patients.

1.3 Research Gap

Although CTG interpretation is critical for monitoring fetal health, existing systems fail to deliver:

- Automated CTG interpretation for home-users
- Image-based CTG graph analysis using mobile phones
- Remote doctor-AI collaboration for final diagnosis
- Instant risk-level classification accessible anytime
- An affordable, non-IoT solution for routine CTG monitoring

Current approaches are insufficient because they rely heavily on:

- Manual doctor interpretation

- Routine hospital visits
- CTG machines without digital/AI support
- Lack of real-time risk prediction

This study introduces a novel contribution: an **AI-assisted CTG analysis module** capable of interpreting both manually entered data and scanned CTG images. It provides evidence-based fetal health insight, supports medical professionals, and enables early detection of complications. This research directly addresses the gap by offering a **mobile, accessible, medically relevant, AI-powered CTG interpretation system** for pregnant women and healthcare providers.

OBJECTIVE

2.1 Main Objective

To develop an AI-powered CTG Interpretation Module capable of analyzing manually entered CTG values or scanned CTG graph images and generating an accurate, clinically meaningful fetal-health assessment for both pregnant mothers and doctors.

2.2 Specific Objectives

The specific goals and objectives of this research are to facilitate pregnant mothers in entering their CTG information, either manually or by capturing images of the CTG printouts and sending them via a mobile device; to develop and execute an AI model that can identify significant features of the CTG, such as variability, accelerations, decelerations, and patterns of uterine contractions; to categorize the health and wellbeing of the fetus into distinct and understandable categories such as Normal, Mild Risk, and High Risk, based on the output of the analysis; to create a detailed and overall summary report that includes abnormalities, possible medical reasons for such abnormalities, and preventive measures that can be taken; to provide doctors with a separate interface for evaluating the output of the AI and making their own comments before submitting the data; and to create a structured history of the CTG data, enabling continuous monitoring and long-term trend analysis for the early detection of possible complications during pregnancy.

3. METHODOLOGY

3.1 Main Objective

This research is based on Design Science Research (DSR) methodology, which emphasizes designing, developing, and evaluating a technological solution for a real-world problem. The key artifact of this research is the AI-based CTG Interpretation Module, which is designed to facilitate remote fetal monitoring by leveraging automation in CTG analysis. The application of DSR methodology is relevant in this context as it emphasizes iterative development, experimentation, and evaluation of the effectiveness of the developed system in solving a real-world problem in healthcare.

3. 2Technical Methods and Algorithms

The proposed system employs a variety of image processing techniques and machine learning algorithms for the interpretation of the CTG data. In the case of image-based data, the pre-processing steps for the image would be noise reduction, conversion of images to grayscale, curve extraction, and segmentation of the patterns of FHR and UC. After pre-processing, a classification algorithm such as Decision Tree, Random Forest, or Lightweight Neural Network is employed to extract features such as variability, accelerations, decelerations, and contractions. In the case of manually entered values of CTG, a statistical approach for feature extraction is employed to calculate the variability indices. These features are passed on to the AI model for classifying the condition of the fetus into a particular risk category

3. 3 Tools and Platforms with Justification

- Sd **Mobile Application (Flutter / Android Studio):** Selected for cross-platform compatibility and smooth integration of camera-based data input.
- **Python & TensorFlow / Scikit-Learn:** Used to build and train the CTG classification model due to their strong support for machine learning algorithms.
- **MySQL / Firebase:** Utilized as the database layer for securely storing CTG data, user details, AI reports, and doctor feedback.
- **OpenCV:** Employed for image preprocessing and CTG curve extraction from scanned images.
  These tools are chosen for their stability, wide adoption in research, and efficient support for medical data processing.

3. 4 Validation Strategy and Performance Matrice

To evaluate the effectiveness of the AI CTG Interpretation Module, several validation techniques will be applied. The model will be tested using **k-fold cross-validation** to ensure generalizability. Performance will be measured using metrics such as **accuracy**, **precision**, **recall**, **F1-score**, and **classification error rate** for Normal, Mild Risk, and High-Risk categories. Image-based CTG analysis will be assessed by comparing AI-identified features with ground-truth annotations. Usability testing will also be conducted to ensure that pregnant mothers and doctors can efficiently interact with the system. The final model performance will be reviewed to ensure reliability suitable for prenatal monitoring support.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4. 1 Overview of the Proposed Solution and Its Components

The proposed solution is a mobile-based pregnancy monitoring system. It includes various intelligent modules to assist pregnant women and doctors. The main idea of the system is the Cardiotocography (CTG) Interpretation Module powered by AI. It allows the monitoring of the well-being of the fetus without the requirement of any equipment. The system includes four main modules:

- User Mobile Application for inputting the data from the CTG and viewing the reports,
- AI CTG Processing Engine for analyzing the values of the CTG or images,
- Doctor Review & Feedback Interface, and
- Cloud Backend & Database for storing the data securely.

This module is the core intelligent layer of the system. It classifies the risks of the fetus using AI. It provides meaningful results for both patients and doctors.

4. 2 System Diagram of Components (Textual Description)

The high-level system architecture functions by following a structured workflow. The workflow begins with the mobile application, where pregnant mothers enter the CTG data. This is done by manually entering the data or by capturing images of the printouts of the CTG. The mobile application sends these entered values to the server, where the AI-powered engine processes the image and analyzes the CTG. The AI engine processes the image, and the CTG is categorized into risk levels such as Normal, Mild Risk, and High Risk. The AI engine then creates a report summarizing these findings. This report is stored on the cloud database and simultaneously sent to the doctor's interface. The doctor reviews the findings generated by AI and adds his own comments, and finally, the findings are sent to the mother. The entered values, reports, and doctor feedback are stored, and historical trend analysis is possible.

4. 3 Highlighting the Individual Component

My component—the **AI CTG Interpretation Module**—consists of four main internal subsystems:

- **CTG Data Intake Unit:** Handles manual CTG value submission and image uploads.
- **Image Preprocessing & Feature Extraction Unit:** Uses filtering, edge detection, and curve segmentation to extract fetal heart rate and contraction signals from CTG images.
- **AI Classification Unit:** Applies machine learning models to classify CTG readings into risk categories.
- **Report Generation Unit:** Converts analysis outputs into clear, readable summaries with explanations and recommended actions.

This module operates as the intelligence core of the entire system, bridging the gap between raw CTG data and meaningful medical insights.

4. 4 Component Interactions

- **User → Mobile App:** The user enters CTG values or uploads a CTG image.
- **Mobile App → AI Engine:** Data is transferred to the server for analysis.
- **AI Engine → Database:** Results, extracted features, and risk levels are stored.
- **AI Engine → Doctor Dashboard:** AI report is sent for professional review.
- **Doctor Dashboard → User App:** Final, doctor-verified report reaches the mother.

My module interacts directly with both the database and the doctor interface, ensuring an integrated decision-support process.

4. 5 Data Flow Explanation

- **Input Stage:**
- CTG values or images are collected.
- **Processing Stage:**
- Images undergo preprocessing (noise removal, segmentation).
- AI extracts CTG features and performs classification.
- **Output Stage:**
- Risk level and summary report are generated.
- **Review Stage:**
- Doctors review the report and add comments.
- **Storage Stage:**
- Final results are saved in the database and made accessible for future reference.

4. 6 Scalability and Security Considerations

The system is designed with scalability in mind, allowing additional AI models, more users, and expanded health-monitoring features to be incorporated without major architectural changes. Cloud-based infrastructure ensures the ability to handle high request volumes and parallel CTG analyses. Security is maintained through encrypted communication (HTTPS/TLS), secure authentication for doctors and mothers, and controlled data-access permissions. All CTG data is stored following medical privacy guidelines to ensure confidentiality and integrity.

5. USER REQUIREMENTS

5. 1 Stakeholder Identification

The system involves several key stakeholders who interact with or benefit from the AI-based CTG Interpretation Module:

- **Pregnant Mothers:** Primary users who upload CTG data and receive AI-generated fetal-health insights.
- **Doctors / Obstetricians:** Medical professionals who review AI reports, provide final comments, and ensure diagnostic accuracy.
- **System Administrators:** Responsible for maintaining backend services, databases, and application performance.
- **Healthcare Institutions (Indirect Stakeholders):** Benefit from improved monitoring quality, reduced workload, and structured reporting.

5. 2 Functional Requirements

1. **The system shall allow pregnant mothers to input CTG data manually** by entering values such as fetal heart rate readings and contraction intervals.
2. **The system allow users to upload CTG images** captured using the mobile device camera.
3. **The system shall preprocess uploaded CTG images** to extract fetal heart rate and contraction curves.
4. **The system shall apply an AI model** to classify CTG readings into risk categories (Normal, Mild Risk, High Risk).
5. **The system shall generate an automated summary report** outlining abnormalities, medical explanations, and recommendations.
6. **The system shall allow doctors to review AI-generated CTG reports** and add professional comments.
7. **The system shall send the final doctor-verified report back to the mother** through the mobile application.
8. **The system shall store all CTG reports in a secure database** for historical tracking and repeated monitoring.
9. **The system shall allow users to view their previous CTG history** and track fetal-health trends over time.
10. The **system shall notify users when a high-risk CTG pattern is detected**, prompting urgent medical attention.

5. 3 Non-Functional Requirements

Performance Requirements

- The system must process CTG images and generate AI results **within 5–10 seconds** to ensure timely feedback.
- The application must maintain stable performance for **hundreds of concurrent users** without degradation.

Usability Requirements

- The user interface must be simple and intuitive for pregnant mothers with varying technical skills.
- The CTG upload and viewing processes must require **minimal steps** to reduce user burden.
- All generated reports must use clear, non-technical language suitable for non-medical users.

Security Requirements

- All CTG data transfers must be encrypted using **HTTPS/TLS**.
- Users and doctors must authenticate securely using passwords or two-factor authentication.
- Only authorized doctors may view or modify CTG interpretation reports.
- The database must follow strict data-privacy practices to protect sensitive medical information.

Reliability Requirements

- The system must maintain **at least 95% up time** to ensure continuous monitoring access.
- Data must be backed up regularly to prevent loss of CTG history.
- The system must correctly store and retrieve CTG reports without corruption or duplication.

Scalability Requirements

- The architecture must support the addition of new AI models and more users without major redesign.
- Storage systems must scale as CTG history grows for each user.

6. COMMERCIALIZATION PLAN

6. 1 Target Market and Customer Persona

The primary target market consists of **pregnant mothers**, particularly those who face difficulties accessing frequent hospital-based CTG evaluations due to distance, cost, or time constraints. Secondary markets include **obstetricians, maternity clinics, private hospitals, and telemedicine platforms** that wish to enhance their prenatal monitoring capabilities. A typical customer persona is a mother in her second or third trimester who requires regular fetal monitoring but prefers remote, AI-supported guidance with doctor validation for faster decision-making.

6. 2 Value Proposition

The AI-based CTG Interpretation Module provides immediate health information for the fetus, eliminating the need for physical hospital visits and enabling timely detection of pregnancy complications. This provides mothers with accessible and accurate risk information, and doctors with a structured and AI-supported reporting framework. This increases the quality of medical care, reduces the medical workload, and ensures safety through continuous monitoring from home.

6. 3 Revenue Model

The proposed business model leverages a **subscription-based structure**, including:

- **Monthly or annual premium plans** for mothers to access unlimited CTG analysis and reports.
- **Clinic integration packages** where hospitals and obstetricians pay a licensing fee to use the doctor-review dashboard.
- **Pay-per-scan model** for occasional users who require only one-time CTG analysis. This mixed model ensures affordability for individuals while maintaining profitability through clinical partnerships.

6. 4 Cost Estimation

stimated initial costs include:

- **AI model development and training** (cloud compute, datasets)
- **Mobile app development** (Flutter/Android)
- **Backend and server hosting**

- **Medical consultation and testing costs**
- **Marketing and deployment**

A realistic early-stage development budget is expected to be around **LKR 3000 – 10000** depending on data acquisition and clinical testing requirements. Ongoing monthly server costs are estimated at **LKR 1000 – 2000** depending on user volume

6. 5 Pricing Strategy

Estimated initial costs include:

- **AI model development and training** (cloud compute, datasets)
- **Mobile app development** (Flutter/Android)
- **Backend and server hosting**
- **Medical consultation and testing costs**
- **Marketing and deployment**

A realistic early-stage development budget is expected to be around **LKR 300,000 – 600,000**, depending on data acquisition and clinical testing requirements. Ongoing monthly server costs are estimated at **LKR 10,000 – 20,000** depending on user volume

6. 6 Competitive Advantage

Compared to existing pregnancy apps that focus mainly on lifestyle tracking and educational content, the proposed system offers:

- **AI-powered clinical CTG interpretation**, not available in mainstream apps
- **Image-based CTG analysis**, allowing mothers to scan paper printouts
- **Doctor-verified results**, increasing medical reliability
- **Remote monitoring capability**, reducing hospital visits
- **Historical CTG trend analysis**, improving long-term fetal health tracking

6. 7 Intellectual Property Considerations

The AI CTG Interpretation Module, including its image-processing pipeline, feature extraction logic, and classification model, may qualify for **software-based intellectual property protection**. The system's unique workflow—combining mother input, AI interpretation, and doctor validation—can be registered as a **proprietary algorithmic process**. Care will be taken to ensure all training datasets are ethically sourced and compliant with medical data regulations to avoid copyright or privacy violations.

7. BUDGET AND JUSTIFICATION

Table 1

#	Item	Estimated Cost (LKR)	Justification
1	Hardware (Mobile device for testing)	15,000	Required for testing CTG image capture, camera quality validation, and app performance on real devices.
2	Software Licenses / Tools	0 – 5,000	Most tools (Python, TensorFlow, Flutter, OpenCV) are free; small optional cost for design tools or premium IDE features.
3	Cloud Services (Database + Hosting)	3,000 – 5,000 per month	Needed for model hosting, secure CTG data storage, user authentication, and running backend AI services.
4	AI Model Training Compute	5,000 – 10,000	Used for GPU/CPU compute time (Google Colab Pro or cloud VM) to train and test CTG classification models efficiently.
5	Human Effort Cost (Development + Documentation)	30,000 – 40,000	Time required for model development, mobile app integration, testing, evaluation, and writing research documentation.
6	Medical Expert Consultation	10,000 – 15,000	Professional doctor input required to validate CTG interpretation, risk-level classification, and final evaluation.
7	Miscellaneous Costs (Internet, printing, transport)	2,000 – 3,000	Used for app testing in different locations, printing research materials, and other small project needs.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Table 2

Task No.	Task Description (Your Component)	Deliverables
1	Gather CTG datasets (numeric + image-based)	Dataset folder, documentation notes
2	Preprocess CTG images (noise removal, segmentation)	Processed images, preprocessing scripts
3	Extract key CTG features (variability, accelerations, decelerations)	Feature extraction module
4	Develop AI classification model for CTG analysis	Trained model + evaluation report
5	Implement risk-level mapping (Normal / Mild / High)	Risk classification logic
6	Build CTG summary report generator	Report template + auto-generated outputs
7	Integrate AI outputs with the doctor review interface	API integration + UI linkage
8	Conduct testing, validation, and performance analysis	Test results, accuracy report
9	Final documentation and module refinement	Final documentation for the proposal

8.2 Timeline Alignment (Your Component Only)

Table 3

Project Phase	Weeks	Your Tasks
Research & Planning	Week 1–2	CTG theory study, dataset collection
Data Preparation	Week 3–4	Image preprocessing + feature extraction
Model Development	Week 5–7	AI model training + tuning
Module Integration	Week 8–9	App + backend integration
Testing & Validation	Week 10–11	Accuracy testing, fixing issues
Finalization	Week 12–14	Documentation, polishing, presentation

8. 3 Balanced Workload Distribution

Table 4

Project Phase	Weeks	Your Tasks
Research & Planning	Week 1–2	CTG theory study, dataset collection
Data Preparation	Week 3–4	Image preprocessing + feature extraction
Model Development	Week 5–7	AI model training + tuning
Module Integration	Week 8–9	App + backend integration
Testing & Validation	Week 10–11	Accuracy testing, fixing issues
Finalization	Week 12–14	Documentation, polishing, presentation

9. GANTT CHART

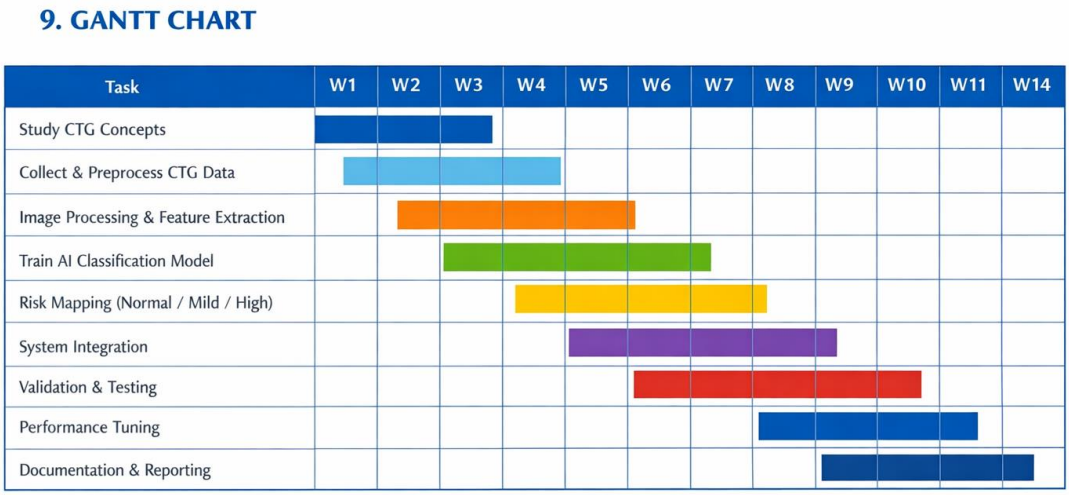


Figure 1

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